Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	1 March 2026
Team ID	GACSE129
Project Name	AI Financial Advisor
Maximum Marks	4 Marks

Technical Architecture:

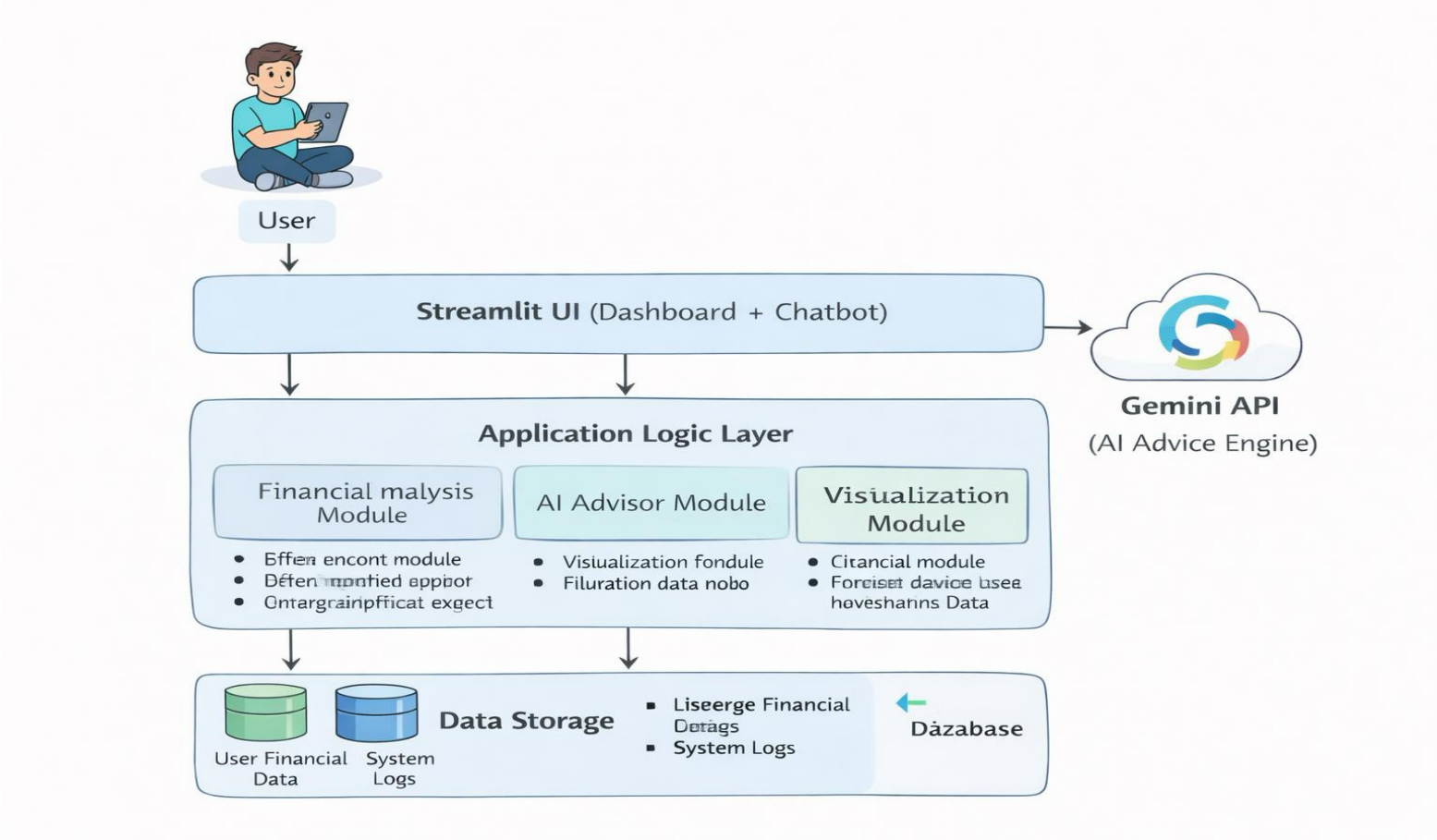

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface for users to input financial data and view insights through dashboard and chatbot	Streamlit Web UI
2.	Application Logic-1	Financial computation module to calculate savings ratio, debt ratio, and emergency fund	Python
3.	Application Logic-2	AI advisory module generating financial guidance and insights	Python + Gemini API
4.	Application Logic-3	Visualization module generating charts and financial reports	Python, Matplotlib, Seaborn
5.	Database	Storage of user financial records and profile information	SQLite / Local Data Storage

6.	Cloud Database	Optional cloud-based storage for scalability	Google Cloud Storage / Firebase
7.	File Storage	Storage of configuration files, logs, and datasets	Local File System
8.	External API-1	AI language model used for generating financial advice	Google Gemini API
9.	External API-2	Optional financial market or economic data APIs	Financial Market APIs
10.	Machine Learning Model	AI model generating personalized financial recommendations	Gemini Large Language Model
11.	Infrastructure (Server / Cloud)	Deployment of application locally or on cloud platform	Local System / Google Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used to build UI, analytics, and AI integration	Streamlit, Pandas, NumPy, Matplotlib
2.	Security Implementations	Secure storage of API keys and protection of user data	.env configuration, API key protection

3.	Scalable Architecture	Modular architecture separating UI, logic, AI services, and visualization	Python Modular Architecture
4.	Availability	System accessible through web interface anytime	Streamlit Web Application
5.	Performance	Efficient data processing and fast AI response generation	Optimized Python processing